

Quantitative Pathway for Protocell Formation Phases

LIFEORIG working note

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1 Purpose

This note defines a quantitative pathway for modeling different physical stages of protocell formation inside the LIFEORIG framework. The aim is not to declare one origin-of-life pathway correct. The aim is to define computable states, transition criteria, and equations that can later be coupled to the atmosphere, hydrosphere, ocean chemistry, geophysical interior, and prebiotic reaction network.

The phases considered here are:

1. porous mineral or ice compartments;
2. solid surface compartments and adsorbed chemical networks;
3. liquid-liquid coacervate compartments;
4. amphiphilic vesicles and true protocell-like compartments.

The working philosophy is:

planetary environment $\rightarrow$ hydrosphere/geochemistry $\rightarrow$ chemical network $\rightarrow$ compartment physics $\rightarrow$ growth/division (1)

2 Environmental Inputs

The protocell module should not invent its own environment. It should receive boundary conditions from the atmosphere, hydrosphere, and geochemistry solvers.

2.1 Atmosphere and hydrosphere inputs

Required environmental variables:

$$T_s \quad \text{surface temperature [K]}, \quad (2)$$

$$P_s \quad \text{surface pressure [Pa]}, \quad (3)$$

$$F_{UV}(\lambda, z) \quad \text{local UV photon flux}, \quad (4)$$

$$\tau_{UV}(\lambda, z) \quad \text{UV optical depth}, \quad (5)$$

$$M_w \quad \text{available liquid/ice water mass}, \quad (6)$$

$$d_w \quad \text{surface water depth or pore water depth}, \quad (7)$$

$$a_w \quad \text{water activity}, \quad (8)$$

$$\Phi_{\text{wetdry}} \quad \text{wet/dry cycling strength}, \quad (9)$$

$$\Phi_{\text{freeze}} \quad \text{freeze/thaw concentration factor}. \quad (10)$$

For a subsurface ocean, use the ocean pressure and temperature at the relevant depth:

$$T_o = T(r_{\text{ocean}}), \quad (11)$$

$$P_o = P(r_{\text{ocean}}). \quad (12)$$

For a surface pond or volcanic-rock environment, use local surface values:

$$T_{\text{env}} = T_s, \quad (13)$$

$$P_{\text{env}} = P_s. \quad (14)$$

2.2 Geochemical inputs

The geochemistry module should provide:

$$\text{pH}, \quad I, \quad \text{Eh}, \quad [\text{Fe}^{2+}], \quad [\text{Fe}^{3+}], \quad [\text{HS}^-], \quad [\text{SO}_4^{2-}], \quad [\text{PO}_4], \quad [\text{Mg}^{2+}], \quad (15)$$

plus mineral availability:

$$S_m(X) \quad \text{reactive surface area of mineral } X. \quad (16)$$

The relevant mineral classes are:

$$X \in \{\text{clay}, \text{FeS}, \text{NiS}, \text{FeOOH}, \text{silica}, \text{phosphate surface}, \text{ice}\}. \quad (17)$$

2.3 Chemical network inputs

The reaction network should provide concentrations or activities of:

$$\text{HCN}, \text{H}_2\text{CO}, \text{NH}_3, \text{CO}, \text{CO}_2, \text{H}_2, \text{H}_2\text{S}, \text{CH}_3\text{SH}, \quad (18)$$

$$\text{formamide}, \text{formate}, \text{acetate}, \text{glyoxylate}, \text{glycolaldehyde}, \text{pyruvate}, \quad (19)$$

$$\text{glycine}, \text{alanine}, \text{aspartate}, \text{glutamate}, \quad (20)$$

$$\text{thioacetate}, \text{acetyl phosphate}, \text{aminoacyl thioesters}, \quad (21)$$

$$\text{peptide oligomers}, \text{hetero oligomers}, \text{amphiphiles}. \quad (22)$$

3 Central Mechanism: Wet-Dry Concentration and Chemical Trapping

The difficult step is not only producing organics. The difficult step is keeping their concentration high enough, for long enough, to build boundaries. Therefore protocell formation should be modeled as a transient environmental cycle, not as a simple average-abundance threshold.

The core sequence is:

$$\text{wet/dry cycle} \rightarrow \text{concentration pulse} \rightarrow \text{amphiphile synthesis/assembly} \rightarrow \text{boundary closure} \rightarrow \text{chemical trapping} \quad (23)$$

3.1 Water-depth controlled concentration

Let $h(t)$ be the local water depth in a pond, pore, film, or wet mineral surface. If the number of molecules n_i is approximately conserved during drying and the wetted area is A_{pond} , then

$$c_i(t) = \frac{n_i(t)}{A_{\text{pond}}h(t)}. \quad (24)$$

If chemistry is slow compared with the drying concentration step, then

$$c_i(t) \propto \frac{1}{h(t)}. \quad (25)$$

This is important because many prebiotic reactions are second order or higher. For example, if an amphiphile precursor L forms by



then

$$\frac{dc_L}{dt} = k_L(T, \text{pH})c_Ac_B - k_{\text{loss}}c_L. \quad (27)$$

If drying increases both c_A and c_B by a factor Γ , the production term increases as Γ^2 .

3.2 Explicit wet-dry concentration term

A minimal local environmental equation is:

$$\frac{dc_i^{\text{ext}}}{dt} = R_i^{\text{env}}(\mathbf{c}, T, \text{pH}) + S_i^{\text{geo}} - k_{\text{wash}}W(t)c_i^{\text{ext}} - \frac{c_i^{\text{ext}}}{h} \frac{dh}{dt}. \quad (28)$$

The last term is the concentration/dilution term. During drying, $dh/dt < 0$, so the term is positive. During rehydration, $dh/dt > 0$, so the term is negative.

A simple periodic water depth can be:

$$h(t) = h_0 + \Delta h \cos\left(\frac{2\pi t}{\tau_{\text{cyc}}}\right), \quad (29)$$

with clipping if needed to avoid negative water depth.

The concentration factor relative to a reference depth h_0 is:

$$\Gamma(t) = \frac{h_0}{h(t)}. \quad (30)$$

3.3 Assembly exposure

The relevant condition is not merely instantaneous threshold crossing. The system must remain above threshold long enough for assembly.

Let c_L be the total concentration of membrane-forming amphiphiles:

$$c_L(r, t) = \sum_{\ell} w_{\ell} c_{L_{\ell}}(r, t). \quad (31)$$

Define the accumulated amphiphile assembly exposure:

$$Q_L(r, t) = \int_0^t dt' \max\left[0, \frac{c_L(r, t') - c_L^{\text{crit}}}{c_L^{\text{crit}}}\right]. \quad (32)$$

Vesicle formation is allowed when:

$$Q_L(r, t) > Q_L^{\text{crit}}. \quad (33)$$

Equivalently, if the concentration is above threshold over a dry interval Δt_{asm} :

$$\Delta t_{\text{asm}} = \int dt \Theta [c_L(t) - c_L^{\text{crit}}], \quad (34)$$

then assembly requires:

$$\Delta t_{\text{asm}} \gtrsim \tau_{\text{asm}}. \quad (35)$$

3.4 Boundary closure traps the chemical state

If a vesicle forms at $t = t_{\text{form}}$, its initial contents should be sampled from the environmental concentration at that time:

$$X_i^k(t_{\text{form}}) \sim \text{Poisson} [c_i^{\text{ext}}(r_k, t_{\text{form}}) V_k^0]. \quad (36)$$

This is the chemical memory of the dry phase. After rehydration, $h(t)$ rises and c_i^{ext} decreases, but the internal contents do not instantly dilute because a boundary exists.

3.5 Leakage after rehydration

For a vesicle, passive exchange is:

$$J_i^k = A_k P_i [c_i^{\text{ext}}(r_k, t) - c_i^{k, \text{int}}(t)]. \quad (37)$$

After rehydration,

$$c_i^{\text{ext}} < c_i^{k, \text{int}}, \quad (38)$$

so $J_i^k < 0$ and molecules leak out.

The leakage timescale is approximately:

$$\tau_{i, \text{leak}} = \frac{V_k}{A_k P_i}. \quad (39)$$

For a spherical vesicle,

$$V_k = \frac{4}{3} \pi R_k^3, \quad (40)$$

$$A_k = 4 \pi R_k^2, \quad (41)$$

so

$$\tau_{i, \text{leak}} = \frac{R_k}{3 P_i}. \quad (42)$$

The trapped enrichment survives if:

$$\tau_{i, \text{leak}} > \tau_{\text{wet}}, \quad (43)$$

or, more generally,

$$\tau_{i, \text{leak}} > \tau_{\text{react}, \text{int}}. \quad (44)$$

3.6 Central timescale window

The wet-dry mechanism works only inside a timescale window:

$$\tau_{\text{synth},L}, \tau_{\text{asm}} < \tau_{\text{dry}} < \tau_{\text{leak}}. \quad (45)$$

If the dry phase is too short, amphiphiles do not form or assemble. If the wet phase is too long relative to leakage, the trapped chemical memory is erased.

4 Compartment Classes

Pores, coacervates, and amphiphile vesicles must be treated as different objects. They are all compartments, but they have different boundary physics.

4.1 Generic compartment

Define a generic compartment k as:

$$\mathcal{C}_k(t) = \left[\Omega_k(t), \partial\Omega_k(t), r_k(t), V_k(t), \{X_i^k(t)\}, B_k \right]. \quad (46)$$

Here Ω_k is the interior region, $\partial\Omega_k$ is the boundary, r_k is the position, V_k is the volume, X_i^k is the molecule number of species i , and B_k is the boundary rule.

4.2 Mineral pore compartment

A pore is a fixed environmental compartment. It is not a free protocell.

$$\mathcal{C}_j^{\text{pore}} = \left[\Omega_j^{\text{pore}}, \partial\Omega_j^{\text{mineral}}, V_j, \{c_i^j(t)\}, \{S_i^{\text{wall}}\}, \{K_{jm}^{(i)}\} \right]. \quad (47)$$

The pore equation is:

$$\frac{dc_i^j}{dt} = R_i^{\text{aq}}(\mathbf{c}^j; T_j, \text{pH}_j, \text{redox}_j) + S_i^{\text{wall}} + \sum_m K_{jm}^{(i)}(c_i^m - c_i^j) + S_i^{\text{geo}} - \frac{c_i^j}{h_j} \frac{dh_j}{dt}. \quad (48)$$

The boundary rule is:

$$\partial\Omega_j^{\text{pore}} = \text{mineral wall}. \quad (49)$$

Pores are initial geometry. They do not divide and they are not born from the chemistry.

4.3 Coacervate droplet compartment

A coacervate is a mobile liquid-liquid phase separated droplet. It is a weak protocell candidate, but it is not a membrane-bounded protocell.

$$\mathcal{C}_k^{\text{coa}} = \left[r_k, R_k, V_k, \phi_k, \{X_i^k\}, \{K_i^k\}, \{\tau_{i,\text{part}}^k\} \right]. \quad (50)$$

Here ϕ_k is the dense-phase volume fraction, K_i^k is the partition coefficient, and $\tau_{i,\text{part}}^k$ is the exchange/partitioning time.

The boundary rule is:

$$\partial\Omega_k^{\text{coa}} = \text{liquid} - \text{liquid interface}. \quad (51)$$

The partition coefficient is:

$$K_i^k = \frac{c_i^{k,\text{in}}}{c_i^{\text{ext}}}. \quad (52)$$

If $K_i^k > 1$, species i is enriched inside the coacervate. If $K_i^k < 1$, it is excluded. Define a local phase-separation score:

$$\Phi_{\text{coa}}(r, t) = \sum_{a \in A} \sum_{b \in B} \alpha_{ab} c_a(r, t) c_b(r, t), \quad (53)$$

where A and B are phase-separating species classes, for example oppositely charged oligomers.

Coacervate formation is allowed when:

$$\Phi_{\text{coa}}(r, t) > \Phi_{\text{coa}}^{\text{crit}}. \quad (54)$$

At formation:

$$X_i^k(t_{\text{form}}) \sim \text{Poisson} \left[K_i^k c_i^{\text{ext}}(r_k, t_{\text{form}}) V_k \right]. \quad (55)$$

After formation:

$$\frac{dc_i^{k,\text{in}}}{dt} = R_i^k(\mathbf{c}^k; T, \text{pH}, \text{redox}) + \frac{K_i^k c_i^{\text{ext}}(r_k, t) - c_i^{k,\text{in}}}{\tau_{i,\text{part}}^k}. \quad (56)$$

A coacervate retains memory of a dry phase only if:

$$\tau_{i,\text{part}}^k > \tau_{\text{wet}}. \quad (57)$$

4.4 Amphiphile vesicle protocell

An amphiphile vesicle is the clean object to call a protocell. It is a mobile membrane-bounded compartment with finite permeability.

$$\mathcal{P}_k^{\text{ves}} = \left[r_k, R_k, V_k, A_k, N_L^k, \{X_i^k\}, \{P_i^k\} \right]. \quad (58)$$

The boundary rule is:

$$\partial\Omega_k^{\text{ves}} = \text{amphiphile membrane}. \quad (59)$$

The membrane exchange equation is:

$$J_i^k = A_k P_i^k \left[c_i^{\text{ext}}(r_k, t) - c_i^{k,\text{int}}(t) \right], \quad (60)$$

with

$$c_i^{k,\text{int}} = \frac{X_i^k}{V_k}. \quad (61)$$

For initial radius R_0 , required amphiphiles are:

$$N_{L,\text{req}} = \frac{4\pi R_0^2}{a_L} \quad (62)$$

for a monolayer or interface coating, and approximately

$$N_{L,\text{req}} = \frac{8\pi R_0^2}{a_L} \quad (63)$$

for a bilayer vesicle.

The membrane coverage is:

$$\theta_k = \frac{N_L^k}{N_{L,\text{req}}^k}. \quad (64)$$

The vesicle is stable if:

$$\theta_k \geq \theta_{\min}. \quad (65)$$

If $\theta_k < \theta_{\min}$, the vesicle is leaky or dissolves.

4.5 Coacervate to vesicle transition

A coacervate can become amphiphile-stabilized by accumulating amphiphiles at its interface:

$$\frac{dN_L^k}{dt} = k_{\text{ads}} A_k c_L^{\text{ext}} - k_{\text{des}} N_L^k + R_L^k. \quad (66)$$

Here R_L^k is internal or interfacial production of amphiphiles. When

$$\theta_k = \frac{N_L^k}{N_{L,\text{req}}^k} \geq 1, \quad (67)$$

the transition is:

$$\mathcal{C}_k^{\text{coa}} \rightarrow \mathcal{P}_k^{\text{ves}}. \quad (68)$$

4.6 Comparison

Object	Boundary	Main rule	Interpretation
Pore	mineral wall	exchange + wall catalysis	environmental reactor
Coacervate	liquid-liquid interface	partitioning K_i	weak protocell
Vesicle	amphiphile membrane	permeability P_i	membrane protocell

5 Phase 1: Porous Mineral or Ice Compartments

5.1 Physical picture

Before true membranes exist, compartments can be provided by pores, cracks, hydrothermal precipitates, pumice-like volcanic rock, clays, or ice eutectic channels. These structures do not require amphiphiles. They concentrate reactants and keep reaction products near catalytic surfaces.

This phase is important because it gives a physical route to:

1. local concentration;
2. surface catalysis;
3. partial isolation;
4. reaction-diffusion gradients;
5. repeated feeding and flushing.

5.2 Pore concentration

Let a pore have radius r_p , length L_p , volume V_p , and surface area A_p :

$$V_p = \pi r_p^2 L_p, \quad (69)$$

$$A_p = 2\pi r_p L_p. \quad (70)$$

The surface-to-volume ratio is

$$\frac{A_p}{V_p} = \frac{2}{r_p}. \quad (71)$$

Small pores strongly enhance surface chemistry because A_p/V_p increases as r_p decreases.

If a dissolved species i is concentrated by evaporation, freezing, or adsorption, write:

$$C_{i,\text{pore}} = \Gamma_{\text{conc}} C_{i,\text{bulk}}, \quad (72)$$

where

$$\Gamma_{\text{conc}} = \Gamma_{\text{wetdry}} \Gamma_{\text{freeze}} \Gamma_{\text{ads}} \Gamma_{\text{flow}}. \quad (73)$$

5.3 Diffusion and residence time

For a species with molecular diffusion coefficient D_i , the diffusion time across a pore of length L_p is

$$t_{\text{diff},i} = \frac{L_p^2}{D_i}. \quad (74)$$

If there is advective flow velocity u , the advective residence time is

$$t_{\text{adv}} = \frac{L_p}{u}. \quad (75)$$

The Damkohler number compares reaction and transport:

$$\text{Da}_i = \frac{t_{\text{transport}}}{t_{\text{reaction},i}}. \quad (76)$$

For a first-order reaction with rate constant k_i :

$$t_{\text{reaction},i} = \frac{1}{k_i}. \quad (77)$$

Chemistry is efficient inside pores when

$$\text{Da}_i \gtrsim 1. \quad (78)$$

5.4 Surface-catalyzed rate

For a mineral-catalyzed reaction



use:

$$R_C = k_T f_{\text{pH}} f_I (1 + \alpha_m S_m) C_A C_B. \quad (80)$$

The temperature factor is:

$$k_T = A_0 \exp\left(-\frac{E_a}{RT}\right). \quad (81)$$

The pH window can be represented by:

$$f_{\text{pH}} = \exp\left[-\left(\frac{\text{pH} - \text{pH}_{\text{opt}}}{\sigma_{\text{pH}}}\right)^2\right]. \quad (82)$$

5.5 Pore compartment viability criterion

A porous compartment is chemically useful if:

$$\Gamma_{\text{conc}} > 1, \quad (83)$$

$$\text{Da}_{\text{key}} \gtrsim 1, \quad (84)$$

$$S_m(X) > S_{\text{min}}, \quad (85)$$

$$t_{\text{res}} > t_{\text{polymerization}}, \quad (86)$$

$$T_{\text{env}} \in [T_{\text{min}}, T_{\text{max}}]. \quad (87)$$

This gives a computable Boolean:

$$\mathcal{P}_{\text{pore}} = \Theta(\Gamma_{\text{conc}} - 1)\Theta(\text{Da}_{\text{key}} - 1)\Theta(S_m - S_{\text{min}})\Theta(t_{\text{res}} - t_{\text{poly}}). \quad (88)$$

6 Phase 2: Solid Compartments and Surface-Bound Networks

6.1 Physical picture

Solid compartments are not closed cells. They are reaction domains where molecules are adsorbed on mineral or ice surfaces. Examples include clay interlayers, FeS/NiS hydrothermal precipitates, phosphate-rich surfaces, silica surfaces, volcanic rock pores, and ice grain boundaries.

This phase can support reaction networks before membranes exist.

6.2 Adsorption equilibrium

For adsorption of species i onto a surface S :



A Langmuir model gives:

$$\theta_i = \frac{K_i C_i}{1 + \sum_j K_j C_j}, \quad (90)$$

where θ_i is fractional surface coverage.

The adsorbed concentration is:

$$C_{i,S} = \theta_i S_{\text{site}}, \quad (91)$$

where S_{site} is the concentration of reactive surface sites.

6.3 Surface polymerization

For monomer addition on a surface:



use:

$$\frac{dC_{P_{n+1},S}}{dt} = k_{\text{surf}}(T, \text{pH}, I)C_{P_n,S}C_M - k_{\text{hyd}}(T, \text{pH})C_{P_{n+1},S}. \quad (93)$$

For peptide-like growth:



where AA^* is an activated amino acid such as an aminoacyl thioester or aminoacyl phosphate.

6.4 Surface retention versus loss

Surface compartments matter only if useful products are retained long enough. For adsorbed polymer P_n :

$$\frac{dC_{P_n,S}}{dt} = R_{\text{form}} - R_{\text{hyd}} - k_{\text{des}}C_{P_n,S} - k_{\text{flush}}C_{P_n,S}. \quad (95)$$

The retention condition is:

$$k_{\text{form}} > k_{\text{hyd}} + k_{\text{des}} + k_{\text{flush}}. \quad (96)$$

7 Phase 3: Coacervate Compartments

7.1 Physical picture

Coacervates are liquid-liquid phase separated droplets. They can form from oppositely charged polymers, peptides, nucleotides, polyions, or mixed organic polymers. They are not lipid membranes, but they can concentrate solutes and create chemically distinct microenvironments.

For LIFEORIG, coacervates are useful as an intermediate between surface-bound chemistry and amphiphilic vesicles.

7.2 Phase separation criterion

Let ϕ be the polymer volume fraction and χ an effective interaction parameter. A simple Flory-Huggins free energy density is:

$$\frac{\Delta G_{\text{mix}}}{k_B T} = \frac{\phi}{N} \ln \phi + (1 - \phi) \ln(1 - \phi) + \chi \phi(1 - \phi), \quad (97)$$

where N is the polymerization degree.

Phase separation is possible when:

$$\frac{\partial^2 \Delta G_{\text{mix}}}{\partial \phi^2} < 0. \quad (98)$$

The spinodal condition is:

$$\frac{1}{N\phi} + \frac{1}{1 - \phi} - 2\chi < 0. \quad (99)$$

Therefore a rough threshold is:

$$\chi > \frac{1}{2} \left[\frac{1}{N\phi} + \frac{1}{1 - \phi} \right]. \quad (100)$$

7.3 Salt and pH dependence

Coacervation depends strongly on ionic strength and charge state. Represent the effective interaction parameter as:

$$\chi_{\text{eff}} = \chi_0 + \chi_{\text{charge}}(\text{pH}) - \chi_I I. \quad (101)$$

Polymer charge can be approximated from acid-base groups:

$$q(\text{pH}) = \sum_a \frac{-1}{1 + 10^{pK_{a,a} - \text{pH}}} + \sum_b \frac{+1}{1 + 10^{\text{pH} - pK_{a,b}}}. \quad (102)$$

Coacervate stability is favored when oppositely charged species coexist:

$$q_1(\text{pH})q_2(\text{pH}) < 0. \quad (103)$$

7.4 Partitioning into droplets

For species i , define a partition coefficient:

$$K_i^{\text{coac}} = \frac{C_{i,\text{droplet}}}{C_{i,\text{bulk}}}. \quad (104)$$

Reaction rates inside coacervates use:

$$C_{i,\text{local}} = K_i^{\text{coac}} C_{i,\text{bulk}}. \quad (105)$$

Thus, for a reaction $A + B \rightarrow C$:

$$R_{C,\text{coac}} = k(T, \text{pH}, I) K_A^{\text{coac}} K_B^{\text{coac}} C_{A,\text{bulk}} C_{B,\text{bulk}}. \quad (106)$$

7.5 Droplet growth

Let R_d be droplet radius. If growth is controlled by uptake of material from solution:

$$\frac{dR_d}{dt} = \frac{D}{R_d} \left(\frac{C_\infty - C_{\text{eq}}}{\rho_d} \right). \quad (107)$$

Coarsening can be represented by a Lifshitz-Slyozov-like law:

$$R_d^3(t) - R_d^3(0) = K_{\text{LSW}} t. \quad (108)$$

7.6 Coacervate viability criterion

Coacervates are viable chemical compartments if:

$$\chi_{\text{eff}} > \chi_{\text{crit}}, \quad (109)$$

$$K_i^{\text{coac}} > 1 \quad \text{for key feedstocks/catalysts}, \quad (110)$$

$$t_{\text{life,droplet}} > t_{\text{reaction}}, \quad (111)$$

$$k_{\text{leak}} < k_{\text{production}} \quad \text{for retained products}. \quad (112)$$

8 Phase 4: Amphiphilic Vesicles and True Protocells

8.1 Physical picture

An amphiphilic protocell is a compartment bounded by a membrane-like assembly: fatty acid vesicles, mixed amphiphile vesicles, lipid-like membranes, or related structures. This is closer to a true protocell because it provides encapsulation, selective permeability, growth, division, and inheritance of internal chemistry.

The key transition is:

$$\text{compartment} + \text{chemical network} + \text{growth/division} \rightarrow \text{protocell}. \quad (113)$$

8.2 Amphiphile production threshold

Let C_A be total amphiphile concentration. Vesicles require:

$$C_A > \text{CMC}, \quad (114)$$

where CMC is the critical micelle/aggregate concentration.

The CMC depends on temperature, pH, ionic strength, and amphiphile chain length:

$$\text{CMC} = \text{CMC}_0 \exp \left[\frac{\Delta G_{\text{agg}}(T, \text{pH}, I, N_c)}{RT} \right]. \quad (115)$$

Here N_c is the hydrocarbon chain length or an equivalent hydrophobicity index.

8.3 Vesicle geometry

For a spherical vesicle of radius R_v :

$$V_v = \frac{4}{3}\pi R_v^3, \quad (116)$$

$$A_v = 4\pi R_v^2. \quad (117)$$

If each amphiphile occupies membrane area a_0 , the number of amphiphiles in one leaflet is:

$$N_{\text{leaflet}} = \frac{4\pi R_v^2}{a_0}. \quad (118)$$

A bilayer vesicle has approximately:

$$N_{\text{amp}} \approx \frac{8\pi R_v^2}{a_0}. \quad (119)$$

8.4 Encapsulation statistics

If a solute has concentration C_i , the expected number of molecules inside a vesicle is:

$$\lambda_i = C_i N_A V_v. \quad (120)$$

The probability that the vesicle contains at least one molecule of i is:

$$P(N_i \geq 1) = 1 - \exp(-\lambda_i). \quad (121)$$

For an autocatalytic set requiring species $i = 1, \dots, m$, assuming independent encapsulation:

$$P_{\text{set}} = \prod_{i=1}^m [1 - \exp(-C_i N_A V_v)]. \quad (122)$$

This connects directly to the idea that autocatalysis requires a sufficient number of molecules. If $C_i N_A V_v \ll 1$, most vesicles do not contain the required species.

8.5 Membrane permeability

For solute i , membrane flux is:

$$J_i = P_i (C_{i,\text{out}} - C_{i,\text{in}}), \quad (123)$$

where P_i is permeability. The concentration change inside a vesicle is:

$$\frac{dC_{i,\text{in}}}{dt} = \frac{A_v}{V_v} P_i (C_{i,\text{out}} - C_{i,\text{in}}) + R_{i,\text{internal}}. \quad (124)$$

Because

$$\frac{A_v}{V_v} = \frac{3}{R_v}, \quad (125)$$

small vesicles exchange molecules faster than large vesicles.

8.6 Osmotic pressure

The osmotic pressure difference is:

$$\Delta\Pi = RT \sum_i (C_{i,\text{in}} - C_{i,\text{out}}). \quad (126)$$

Membrane tension due to pressure difference can be estimated by Laplace's law:

$$\Delta P = \frac{2\gamma}{R_v}, \quad (127)$$

where γ is membrane tension. Vesicle rupture or division can occur when γ exceeds a critical tension γ_c .

8.7 Growth

If amphiphiles are supplied from the environment, membrane area grows as:

$$\frac{dA_v}{dt} = a_0 J_A A_v, \quad (128)$$

where J_A is the net amphiphile insertion flux per unit membrane area.

For spherical vesicles:

$$\frac{dR_v}{dt} = \frac{a_0 J_A R_v}{2}. \quad (129)$$

Internal chemistry grows if:

$$\frac{dN_{\text{cat}}}{dt} = R_{\text{cat}} V_v - k_{\text{loss}} N_{\text{cat}} > 0. \quad (130)$$

8.8 Division

A simple division criterion can be based on surface-to-volume imbalance. If membrane area grows faster than volume:

$$\frac{dA_v/A_v}{dt} > \frac{2}{3} \frac{dV_v/V_v}{dt}, \quad (131)$$

the vesicle becomes geometrically unstable.

A phenomenological division rate can be written:

$$k_{\text{div}} = k_0 \Theta(A_v - A_{\text{crit}}) \Theta(\gamma - \gamma_{\text{crit}}). \quad (132)$$

9 Autocatalysis and Catalysis Probability

9.1 Molecule-number criterion

For a catalytic polymer species P , the expected copy number in a compartment of volume V_c is:

$$\lambda_P = C_P N_A V_c. \quad (133)$$

The probability that the compartment contains at least $N_{\min}$ copies is:

$$P(N_P \geq N_{\min}) = 1 - \sum_{n=0}^{N_{\min}-1} \frac{\lambda_P^n e^{-\lambda_P}}{n!}. \quad (134)$$

9.2 Catalysis probability

Let a polymer of length n have probability

$$p_{\text{cat}}(n, c) \quad (135)$$

of catalyzing reaction class c . A first parameterization is:

$$p_{\text{cat}}(n, c) = p_{0,c} \left[1 - \exp \left(-\frac{n - n_{\min,c}}{n_{0,c}} \right) \right] \Theta(n - n_{\min,c}). \quad (136)$$

The effective catalyzed rate for reaction r is:

$$R_r = R_{r,0} \left[1 + \sum_P \alpha_{rP} p_{\text{cat}}(n_P, c_r) N_P \right]. \quad (137)$$

9.3 Autocatalytic closure criterion

A reaction set is autocatalytic if there exists a subset $\mathcal{S}$ such that:

1. every reaction in $\mathcal{S}$ is catalyzed by at least one molecule produced by $\mathcal{S}$;
2. every reactant can be made from the food set plus products of $\mathcal{S}$.

Computationally, define a catalysis matrix:

$$K_{ij} = \begin{cases} 1, & \text{species } i \text{ catalyzes reaction } j, \\ 0, & \text{otherwise.} \end{cases} \quad (138)$$

Define stoichiometry matrix ν_{ij} :

$$\frac{dC_i}{dt} = \sum_j \nu_{ij} R_j(C, T, \text{pH}, I). \quad (139)$$

Autocatalytic growth is detected when a positive eigenvalue appears in the linearized production operator around the feedstock state:

$$\max \text{Re} \left[\lambda \left(\frac{\partial \dot{C}}{\partial C} \right) \right] > 0. \quad (140)$$

10 Transitions Between Phases

10.1 Pores to coacervates

The transition from pore chemistry to coacervates requires sufficient polymer production and phase separation:

$$C_{\text{poly}} > C_{\text{LLPS}}, \quad (141)$$

$$\chi_{\text{eff}} > \chi_{\text{crit}}, \quad (142)$$

$$K_i^{\text{coac}} > 1 \quad \text{for catalytic species.} \quad (143)$$

10.2 Pores or coacervates to amphiphilic vesicles

The transition to vesicles requires amphiphile production:

$$C_A > \text{CMC}, \quad (144)$$

$$T_{\text{env}} \in [T_{\text{mem,min}}, T_{\text{mem,max}}], \quad (145)$$

$$\text{pH} \in [\text{pH}_{\text{mem,min}}, \text{pH}_{\text{mem,max}}], \quad (146)$$

$$I < I_{\text{destabilize}}. \quad (147)$$

10.3 Vesicle to protocell

A vesicle becomes protocell-like when it couples internal chemistry to compartment growth:

$$P_{\text{set}} > P_{\text{min}}, \quad (148)$$

$$\frac{dN_{\text{cat}}}{dt} > 0, \quad (149)$$

$$\frac{dA_v}{dt} > 0, \quad (150)$$

$$k_{\text{div}} > 0, \quad (151)$$

$$\text{heritability} > h_{\text{min}}. \quad (152)$$

Here heritability can be defined operationally as the correlation between parent and daughter catalytic composition:

$$h = \text{corr}(\mathbf{N}_{\text{cat,parent}}, \mathbf{N}_{\text{cat,daughter}}). \quad (153)$$

11 Minimal Dynamical Model

For each compartment type c , evolve:

$$\frac{dC_i^{(c)}}{dt} = \sum_j \nu_{ij} R_j^{(c)} + \frac{A_c}{V_c} P_i^{(c)} (C_i^{\text{env}} - C_i^{(c)}) - k_{\text{loss},i}^{(c)} C_i^{(c)}, \quad (154)$$

$$\frac{dV_c}{dt} = G_c(C, T, \text{pH}, I, a_w), \quad (155)$$

$$\frac{dN_c}{dt} = k_{\text{birth}}^{(c)} N_c - k_{\text{death}}^{(c)} N_c. \quad (156)$$

For pores, V_c is usually fixed. For coacervates, V_c changes by phase separation and coarsening. For vesicles, V_c and membrane area evolve through amphiphile uptake, osmotic pressure, and division.

12 Implementation Pathway

12.1 Step 1: define a ProtocellEnvironmentState

The state should contain:

$$T, P, \text{pH}, I, a_w, F_{\text{UV}}, \Phi_{\text{wetdry}}, \Phi_{\text{freeze}}, S_m(X), C_i. \quad (157)$$

12.2 Step 2: evaluate compartment availability

Compute:

$$\mathcal{P}_{\text{pore}}, \quad \mathcal{P}_{\text{surface}}, \quad \mathcal{P}_{\text{coac}}, \quad \mathcal{P}_{\text{vesicle}}. \quad (158)$$

These are not final probabilities at first. They can start as Boolean viability flags.

12.3 Step 3: run chemistry inside each viable compartment

For each viable compartment:

$$\frac{d\mathbf{C}^{(c)}}{dt} = \mathbf{SR}^{(c)} + \mathbf{F}_{\text{exchange}}^{(c)} - \mathbf{L}^{(c)}. \quad (159)$$

12.4 Step 4: test autocatalytic threshold

Track:

$$N_{\text{polymer}}, \quad P_{\text{set}}, \quad p_{\text{cat}}, \quad \max \text{Re}(\lambda). \quad (160)$$

The first computational success condition is not “life”. It is:

$$\text{persistent compartmentalized autocatalytic chemistry}. \quad (161)$$

13 Reference Starting Points

The following literature is useful for model construction:

- Deamer and Dworkin: chemistry and physics of amphiphilic compartments and prebiotic membranes.
- Szostak and collaborators: fatty-acid vesicles, protocell growth, membrane permeability, and encapsulation.
- Hanczyc, Fujikawa, and Szostak: experimental growth and division of fatty-acid vesicles.
- Monnard and Deamer: membrane self-assembly under prebiotic conditions.
- Budin and Szostak: physical effects of membrane composition and competition between protocells.
- Koga, Williams, Perriman, and Mann: protocell models based on coacervates and phase-separated droplets.

- Dora Tang and collaborators: coacervates, protocell droplets, and biochemical compartmentalization.
- Martin, Russell, and collaborators: alkaline hydrothermal vents, mineral compartments, and geochemical energy gradients.
- Kauffman: autocatalytic sets and collective autocatalysis.
- Hordijk and Steel: RAF theory for autocatalytic reaction networks.

14 Working Interpretation

The sequence to test in LIFEORIG should be:

pores/surfaces $\rightarrow$ polymer enrichment $\rightarrow$ coacervate or vesicle compartment $\rightarrow$ autocatalytic chemistry $\rightarrow$ growth (162)

For surface volcanic environments, wet/dry pore and vesicle pathways are probably most relevant. For hydrothermal vents, mineral pore and surface-bound networks are the natural first compartment. For icy ocean worlds, eutectic ice channels and hydrothermal pore compartments may be more plausible than surface pond vesicles, unless there is local exposure, brine cycling, or impact/thermal processing.